

Rule-Guided Reinforcement Learning for German Text Simplification

Kristina Richert

IM - Project Paper SS25

[GitUp Code Repo](#)

Student-ID - 824264

kristina.richert@uni-potsdam.de

1 Abstract

Current LLM-based approaches to German language simplification often struggle to strictly adhere to the guidelines of simplified language. This study investigates to what extent an explicit, two-stage rule-guided methodology can enhance the quality of German text simplification. I first developed a deterministic pipeline to improve guideline adherence on sentences already simplified to a simplified language level (CEFR A2). Subsequently, I use the encoded rules as a numeric reward signal in an RLHF framework to determine if this approach can produce outputs that are both simpler and more guideline-compliant, while preserving semantic integrity. The findings demonstrate that while a customized RLHF process can steer a language model toward greater rule adherence, it simultaneously reveals a critical tension between rule compliance and the preservation of semantic meaning. This trade-off is particularly evident in the underlying deterministic pipeline, which, while effective on simple sentences, can degrade meaning in more complex grammatical structures. Furthermore, qualitative analysis highlights common RLHF challenges, including reward hacking and the generation of undesirable artifacts. Ultimately, this study validates rule-guided RLHF as a promising paradigm for enforcing linguistic constraints but reveals that the primary obstacle remains the development of a reward function that can holistically capture the multifaceted quality of text simplification.

2 Introduction

Text Simplification (TS) describes a task within the Natural Language Processing (NLP) domain that involves modifying the original textual structure in an easier and more straightforward manner while maintaining the contextual integrity of the content. This task has the goal of enabling accessibility and

understanding of complex textual content for a specific group who would greatly benefit from this. It could benefit readers facing a range of challenges, such as individuals with linguistic or cognitive impairments or those who are learning a second language (Alva-Manchego et al., 2020).

Simplified language has gained attention and a movement towards a standardized framework of this concept in the early 2000s. Four recurring notions can be associated with simplified language - *clarity*, *easiness*, *plainness*, and *simplicity* (Vecchiato, 2022).

The field of Automatic Text Simplification (ATS), which involves the automatic production of a simplified text, is a widely growing field which has grown increasingly in recent years (Stajner, 2021). Historically speaking, text simplification was a manual, and therefore time-consuming and expensive process. However, the development in computing power, the availability of high-quality linguistic resources have enabled the possibility in automating this task (Rennes and Jönsson, 2015). Development in this field would also greatly improve the performance within different NLP tasks, such as parsing, text summarization or machine translation (Alva-Manchego et al., 2020). Looking at the development across languages, ATS pipelines have been developed in for English, Portuguese, French, Spanish, Swedish. These approaches span for example, early rule-based (Chandrasekar et al., 1996; Aluísio and Gasperin, 2010) and machine-translation-based models (Coster and Kauchak, 2011) as well as more recent neural and Large Language Model (LLM) based methods. This modern approach leverages large, pre-trained transformer architecture and typically falls into two main directions: 1) fine-tuning pre-trained models on specific text simplification datasets (Hewett et al., 2024), and 2) using large-scaled instruction-tuned models to perform simpli-

simplification through carefully crafted prompts¹.

While the field of ATS has shown a growth in interest and great distributions, the research remains at an early stage. A recent study has shown that, while the field shows growing development, German corpora remain limited in size, applied rules are scattered and not unified and that there is still great room to further develop existing tools (Madina et al., 2023). Stodden et al., 2023 states that "the success of automatic text simplification systems is highly dependent on the quality of parallel data used for training and evaluation". The development of such corpora like DEPlain (Stodden et al., 2023) and Klexikon (Aumiller and Gertz, 2022) show crucial development in the right direction. On another note, with regards to the existent resource scarcity in German, other research has contributed to the field relying on synthetic generated data (Anschütz et al., 2025; Klöser et al., 2024).

Despite these recent advances, current LLMs are not sufficiently aligned with the guidelines and rules of simplified language. One study, evaluating ChatGPT and the expected structure of easy language (*Leichte Sprache*), found that while the outputs are more readable, they frequently violate rule standards and sometimes introduce content errors (Deilen et al., 2023). While LLMs continue to surprise with their level of performance, their behavior remains *opaque* or also described as *black-box* which is a known limitation of LLMs. As a result, there is a clear need for approaches leveraging the strength of LLMs while enforcing explicit rule compliance.

Reinforcement Learning with Human Feedback (RLHF) has recently emerged as a new approach to align LLMs with external objectives (Ouyang et al., 2022; Christiano et al., 2023). RLHF introduces a reward model trained to represent task-specific preferences or constraints, and optimizes the policy model through reinforcement learning (often Proximal Policy Optimization, PPO).

To address this gap, this paper introduces a novel, two-stage methodological approach for aligning LLMs with explicit simplification rules. The process begins by focusing on enhancing existing simplifications rather than simplifying complex texts from scratch. First, a deterministic, rule-guided text simplification system is applied to texts that have already been simplified (CEFR A2), with the objec-

tive of generating a more stringently rule-compliant version of each sentence. In the second stage, these deterministic transformations are quantified as numeric reward signals for a Reinforcement Learning (RL) framework. These rewards, which encode adherence to specific linguistic rules, are then used to optimize a fine-tuned LLM via PPO. The core hypothesis is that by aligning an LLM with rewards derived directly from explicit simplification rules, the model can learn to produce outputs that conform more strictly to simplified language guidelines than is achievable through standard fine-tuning or prompting alone. However, this study's findings reveal a critical trade-off: while the rule-based pipeline is effective on simple sentences, it can degrade meaning in complex ones. Furthermore, while the subsequent reinforcement learning shows promise, the model exhibits common RLHF challenges such as reward hacking and hallucinations. This underscores the core challenge of crafting a reward function that can robustly measure the multifaceted quality of text simplification.

This paper is organized to reflect the two-stage approach. Following the introduction, Section 3 reviews the background of text simplification and RLHF. Sections 4 and 5 are dedicated to the first stage: the design and evaluation of our deterministic, rule-based simplification pipeline. Section 6 details the second stage, presenting the experimental framework, training, and results of the rule-guided reinforcement learning pipeline. A comprehensive discussion of the findings from both stages is presented in Section 7.

3 Background

3.1 Simplified Language: Guidelines and Distinctions

Simplified language aims to make knowledge in textual form more accessible to groups of people facing challenges in reading and understanding difficult structures, such as those with cognitive impairments, low literary or second-language learners. Two distinctions are available within the German context: *Leichte Sprache* (Easy Language) and *Einfache Sprache* (Plain Language). Easy languages can be defined as a firmly-rule based variety with strict guidelines concerning vocabulary, syntax and layout whereas plain language is less standardized describing an "in-between" state of easy language and simplified forms of standard German (Maaß, 2020).

¹<https://github.com/machinelearningZH/simplify-simplify-language>

Several guidelines for simplified language in German exist; however, Maaß’s work stands out for it clearly systematic and linguistically motivated framework. These rules are grouped at the following levels: character, word, sentence, text, and layout (Suter et al., 2016). Key examples include:

- **Character-level:** limitation of allowed characters, replacement of written numbers with digits, visual segmentation of compounds using the *Mediopunkt*.
- **Word-level:** restriction to common and simple words, avoidance of foreign or technical words, unless explained
- **Sentence-level:** one piece of information per sentence, active voice, present or perfect tense
- **Text-level:** preference for repetition over stylistic variety, direct speech over indirect, insertion of explanations or examples where applicable
- **Layout:** large font, one sentence per line, clear segmentation.

3.2 Reinforcement Learning with Human Feedback (RLHF)

Reinforcement Learning with Human Feedback (RLHF) has recently emerged as a new way to align LLMs with human-defined goals. Instead of relying exclusively on supervised-fine tuning, RLHF combines three components: (1) a policy model that will be trained, (2) a reward model that scores output based on the task-specific preferences and (3) a reinforcement learning algorithm that optimizes the policy model to maximize the reward.

Christiano et al., 2023 demonstrated RLHF for complex tasks by training a reward model on human preference comparisons without utilizing the reward function. Ouyang et al., 2022 extended the method to align LLMs with instructions and user intent leading to the success of instruction-following systems such as InstructGPT.

These rules Maaß, 2020, 2015 and the research conducted by Suter et al., 2016 form the linguistic foundation of my work of translating the rules into a deterministic rule-based simplification pipeline. In the second step, I adapt the RLHF approach by substituting human-labeled feedback with rule-based feedback. The simplification rules are incorporated into the reward function, combining rule compliance, semantic preservation and grammatical correctness.

4 Data and Resources

4.1 Data

In this project, I used the DEplain-APA corpus which is a dataset containing parallel corpora of professionally written and manually aligned simplifications in plain German (Stodden et al., 2023). It is a dataset designed to advance research in sentence- and document-level text simplification. The corpus contains Austrian texts that were provided by the Austrian Press Agency (APA) and are within the news domain.² The corpus provides the language level from the original sentence and the target sentence (simplification) in CEFR levels B1 to A2, respectively. Overall, the corpus contains the following aligned structure: 483 document and 13,122 sentence pairs with a ratio of 25,607 complex and 16,471 simplified sentences. For this project, I worked with the sentence aligned format. For this study, the simplified sentences (target level A2) from the DEplain-APA corpus were used as the primary input for the rule-based pipeline and are described as ‘original’ throughout the paper.

4.2 Resources

For linguistic preprocessing, I relied on several resources. Tokenization, part-of-speech tagging, and syntactic dependency parsing were performed with the spaCy library, using the de_core_news_lg model. To improve sentence segmentation for texts containing headlines or list structures, as usually found in the news domain, I integrated a custom rule-based component, newline_sentencizer, into the spaCy pipeline. This component enforces sentence boundaries at newline characters (\n) and is executed prior to the main parser.

For compound segmentation, I employed the German compound splitter³, supplemented with a German dictionary containing more than 2.1 million entries⁴ as recommended by the developers. To handle abbreviations, a resource was constructed from Wikipedia, comprising 405 abbreviations and 278 acronyms⁵. Verb conjugation was addressed

²Datasource can be downloaded here. Request is required:<https://zenodo.org/records/7674560>

³https://github.com/repodiac/german_compound_splitter

⁴<https://sourceforge.net/projects/germandict/files/latest/download>

⁵https://de.wikipedia.org/wiki/Portal:Abkürzungen/Gebräuchliche_Abkürzungen, last accessed May 25th

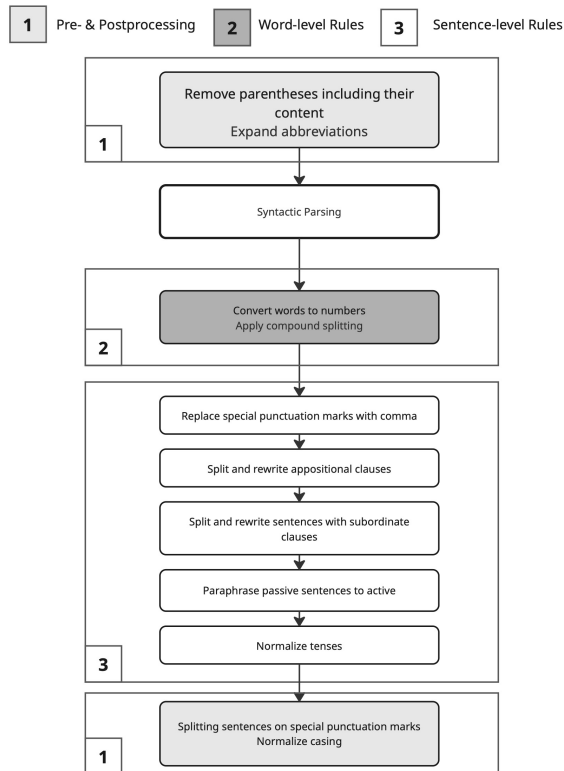


Figure 1: Architecture of the deterministic rule-based simplification pipeline. The process is organized into three main stages: (1) pre- and postprocessing, (2) word-level rules, and (3) sentence-level rules. Adapted from Suter et al., 2016

using the pattern library⁶, together with a German verbs database⁷ to determine auxiliary verbs required in compound tense formation.

5 Rule-based Simplification Pipeline

5.1 Applied Method

This chapter describes the implemented step within the deterministic rule-based simplification pipeline. This work builds upon the research and findings by Suter et al., 2016 and the guideline-defining work by Maaß, 2015, 2020. The steps of coordinate simplification and the switch of complicated words with their alternative counterpart were initially attempted but then excluded from the process.

5.2 Text Processing & Syntactic Parsing

This section is portrayed in step 1 in Figure 1. To prepare the source text for the simplification pipeline, it first undergoes multi-step pre-

processing stages to normalize the input and to structure the data for subsequent parsing. The process is divided into two main components: the initial text normalization and syntactic parsing. First, parentheses including their content are removed entirely from the text, as it is considered to often contain supplementary information which is not crucial for the main sentence. Following this, abbreviations are matched and expanded to their full forms using a manually curated list. Finally, character-level normalization is performed. First, specific characters from a pre-defined list are substituted with their word equivalents and then any remaining character not conforming to a defined set of German letters, numbers and basic punctuation are removed. This stage is then finalized by the parsing and tokenization step. Once sentence-level simplifications have been applied, the pipeline concludes after two more rules. First, sentence splitting is applied if the following characters are found - such as semicolons, colons, and dashes, as long as not identified within a compound structure. Then, a final casing normalization is applied on the sentences.

5.3 Word-level rules

As a next step, the simplification pipeline performs two key transformations on a word-level. Numbers that are written as words are replaced by their equivalent number (with the exception of the indefinite article "ein" ('a')). Nouns are assessed whether they are compounds if they are not named entities (such as personas, locations or organizations) and if they contain more than five characters. If this is proven true, then the compounds are split using the *Mediopunkt*. In cases where compounds are already hyphenated, the hyphen is exchanged with a *Mediopunkt*.

5.4 Sentence-level rules

Following the initial **word-level** transformations, the pipeline executes a sequence of rules aimed at simplifying syntactic structures on a sentence level. These rules are applied iteratively to the output of the preceding step.

If detected, **appositions** are transformed into two independent sentences, turning the complex noun phrasal structure into a more direct subject-verb-object shape. Sentences containing subordinate clauses are identified based on the presence of subordinate conjunctions.

A **subordinate clause** is identified by detecting

⁶<https://digiasset.org/html/pattern-de.html>

⁷<https://github.com/viorelsfetea/german-verbs-database>

predefined marker words such as subordinate conjunctions. If detected, the clause is split into two standalone sentences, the main and subordinate clause. Both are then transformed into two standalone sentences.

Upon detection, **passive voice** construction is converted into the active voice. If an grammatical agent is detected within the sentence starting with (*von ...* the object of the passive sentence), it is transformed into the the subject of the new sentence with an active voice. If no agent is found, the impersonal pronoun *man* ('someone'/'one') is instead introduced as the subject.

In the last step, **tense normalization** is applied where disallowed tenses are normalized into a simpler tense, handling both verb and auxiliary. The following transformations are applied: Future tense is converted into the present tense. Verbs in the simple past or subjunctive are transformed into present perfect. First, the verb's past participle is identified and the correct auxiliary verb is chosen by matching it with a predefined verb database (cf. 4.2). The new sentence is then reconstructed by placing the conjugated auxiliary verb after the subject and moving the past participle to the end of the clause.

5.5 Evaluation

The goal of evaluating automatically simplified texts is to determine whether the output portrays a version which is easier to understand while remaining grammatically correct and maintaining the original meaning. Two of the most traditional methods are readability formulas such as LIX ('Läsbarhetsindex') (Björnsson, 1968) and FLESCH (Flesch, 1948). LIX combines two factors in its calculation: the average number of words per sentence and the percentage of long words (more than 6 letters) (Lenhard and Lenhard, 2011). Flesch measures textual difficulty, with higher scores indicating easier-to-read text. The formula is based on the average number of syllables per word and words per sentence.

However, given that this study's pipeline aims to enforce specific rules on already simplified sentences, such evaluation metrics would not be suitable. An already simplified sentence would likely receive a high score, failing to capture the success of failure of specific rule applications. Thus, the evaluation framework is designed to answer two more targeted questions: 1) Is the meaning of the input sentence preserved after the rule application? and 2) Was the linguistic rule applied successfully

and correctly?

To address the first question, the rule-based simplification was assessed by evaluating the meaning preservation by quantifying the semantic similarity between input and output on a token level using BERTScore (Prashanth, 2025; Zhang et al., 2020). A high score indicates that the transformation did not significantly alter the original meaning (cf. Appendix A further details).

This quantitative measure is complemented by a manual qualitative analysis to address the second question. For each rule, a selection of 2-4 output examples were inspected to determine how well the transformation was executed, identify common error patterns, and assess the overall quality of the simplification (cf. Table 4).

5.5.1 Quantitative Analysis

Table 3 reports the average BERTScore-F1 values for each simplification rule, as an attempt to quantify semantic preservation. To mitigate the issue of rule co-occurrence, I exploded the dataset such that each rule application was evaluated independently, ensuring that the reported BERTScore values reflect per-rule effects rather than bundled transformations. Higher F1 values indicate closer semantic similarity between original and simplified sentences. The results can be clustered in three bands: a high score (> 0.97) can be observed with the rules apposition and tense normalization, a mid-band ($0.95-0.97$) for compound splitting and word to number conversion and a lower-band (< 0.95) for passive to active and subordinate conversion showing greater risk of semantic loss. While useful as a diagnostic, this evaluation comes with important limitations:

- False positives occurred, particularly in tense normalization or apposition, where, despite an applied tag, the input and output could identical leading to a high score
- BERTScore focuses on semantic similarity but does not capture other factors, such as readability or grammatical correctness. Therefore, the table should be understood as a first quantitative approximation rather than a definitive measure of simplification quality.

5.5.2 Qualitative Analysis

Split Compound. Compound segmentation generally improves readability by visually separating long words, as seen in successful applications like

"Vize-Kanzler" and "Corona-Test". However, the dictionary-based approach of the splitter has notable limitations. It occasionally produces unnatural splits by matching sub-strings to valid existing dictionary entries, resulting in examples such as: *Verb-Rechen*, *Übern-Achtungen*, *Bergs-Teigen*. These instances of over-segmentation occur because the component words are technically present in the used German dictionary, leading to correct but contextual incomprehensible splits.

Convert Word to Number. This rule is applied effectively in most cases, correctly converting the written out number to a digit. However, one problem was observed there, in cases of ordinal numbers the phrase "...zum 5. Mal..." becomes "...zum 5 Mal..." ('for the fifth time'). This results in a critical change of meaning from 'the fifth time' to 'five times'.

Normalize Verb Tense. This rule reliably replaces simple past ('Präteritum') forms with present perfect ('Perfekt') in simple declarative sentences, which is the most common tense simplification required in the dataset. Nevertheless, failures were observed in several cases. The rule sometimes fails to correctly identify the grammatical case of the subject, resulting in subject-verb agreement errors (singular/plural mismatch) (Table 5, Ex. 4). Another observed issue is the generation of an ungrammatical word word order. In example 3, the participle is incorrectly placed after the subordinate clause. On another note, false positives were observed, where the simplification output remains the same as the input. For instance, in sentences containing modal verbs, where an infinitive verb at the end of a clause was likely incorrectly flagged for normalization without the system recognizing its syntactic role within the larger modal construction.

Rewrite Apposition. Within simple constructions, the rule successfully transforms appositions into independent sentences while preserving the original referential detail and structure (Table 5, Ex. 1). However, the rule struggles significantly with more complex sentence structures. I identified a noticeable error where a list enumeration was wrongly identified as an appositional phrase. In one example (Table 5, Ex. 2), *ihren Bruder* ('her brother') was incorrectly identified as an appositive renaming *seine ehemalige Freundin* ('his former girlfriend'). This led to the semantically illogi-

cal but also grammatically flawed output "seine ehemalige Freundin ist ihren Bruder" ('his former girlfriend is her brother'), which also contains a grammatical case error.

Simplify Subordinate. This rule aims to split sentences by converting the subordinate clause into a main clause and replacing the original conjunction (e.g. "weil") with an adverb (e.g. "deshalb") to preserve the logical connection. The core logic, splitting the sentence and replacing the conjunction, shows that the intended structural rewriting is partially successful. In example 2, for instance, the successful substitution of "obwohl" with the corresponding adverb "trotzdem" demonstrates that the high-level logic works. However, the applied changes are undermined by occurring mistakes, such as retaining word order which outputs grammatical faulty phrases as seen in the examples in table 5.

This outcome highlights a key limitation of the pipeline: a dependency on the output of the previously applied rules. For instance, in example 5, it can be observed that a flawed output by the tense normalization led to the case where "floss" ('flowed') was incorrectly conjugated (using "hat" instead of the correct auxiliary "ist").

I acknowledge that subordinate simplification yielded some degraded outputs. However, given its low frequency in comparison to other rules and topic coverage importance (as one of the key rule areas), I decided to retain it while flagging it for future refinement.

Convert Passive to Active. Simple passive structures with explicit or clearly defined and easily inferred agents are successfully rephrased into active voice. In these cases, the agent is correctly identified and rephrased as the subject of the new active sentence. The rule's weakness, however, becomes visible in complex sentence structures with embedded clauses. Such constructions often lead to the loss of referential nuance and information. Example 3 in table 5 displays such an occurrence demonstrating a failure to robustly identify the correct agent and subject roles, which in turn lead to a significant semantic drift.

6 Reinforcement Learning Pipeline

6.1 Experimental Framework

Figure 2 portrays the experimental framework of the implemented RLHF pipeline and is described

in the following sections.

Deterministic Simplification. As portrayed in section 1 each original sentence is first simplified using the rule-based pipeline. This data-pair (original, simplified) together with the applied rules per sentence is the required data and information required for the next steps.

Reward Model Training. In the next step, the simplifications are fed into the reward function and scalar rewards are computed from three main components, further described in 6.3. These scores are the baseline to train the reward model (cf. Figure 2, section 2).

PPO Training. Figure 2, section 3 displays the core of the framework. In the reinforcement learning stage, the policy model π_{PPO} is initialized together with a reference model π_{ref} , which represents a frozen copy of the initial fine-tuned checkpoint. The reference model acts like a stable baseline whereas the policy model is learning and iteratively updated. For every simplification generated by the policy model, two steps happen within the training process.

First, the trained reward model calculates a scalar score

$$r_{\phi}(y | x)$$

for the generated simplification y given input x , where ϕ are the parameters of the reward model. To prevent the policy from drifting too far away from the pretrained model state, an additional prediction shift penalty is applied:

$$-\lambda_{\text{KL}} D_{\text{KL}}(\pi_{\text{PPO}}(y | x) \| \pi_{\text{ref}}(y | x))$$

The Kullback–Leibler (KL) divergence penalty is a regularization technique used to ensure training stability. It measures how much the policy model’s output distribution has "diverged" from the original reference model’s distribution. By penalizing large deviations, it prevents the policy from generating nonsensical or repetitive text simply to "hack" the reward model, ensuring it retains its core language capabilities. The final reward passed to the PPO update is thus the difference of the calculated reward score and the penalty.⁸

Evaluation. The trained PPO is then evaluated on a held-out dataset.

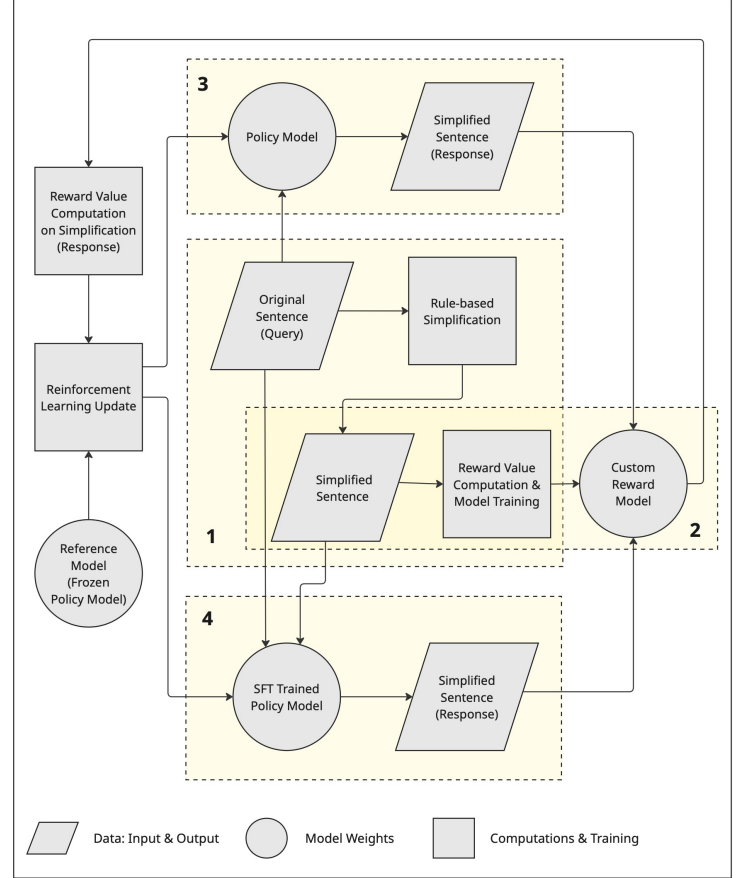


Figure 2: Pipeline overview of the applied RLHF Training setup. The architecture is structured into 4 stages: 1) Deterministic Simplification, 2) Reward Model Training, 3) PPO Training, and 4) Expand Framework with SFT

6.2 Expand Framework with SFT

The second experimental approach expands the framework by inserting a Supervised Fine-Tuning (SFT) stage prior to PPO training (Figure 2, section 4). In this step, the model is further fine-tuned on an instruction-style input derived from the rule-based pipeline’s output.

Each input is formatted as a task-specific instruction including context, presenting the model with the original sentence and the corresponding deterministically simplified version as the target response.

The resulting model from this SFT phase is then used to initialize both, the policy model and the frozen model for the following PPO training. This approach ensures that the reinforcement learning stage begins with a policy already familiar at the specific format and task.

⁸<https://huggingface.co/blog/rlhf>

6.3 Transformation of Rules into Reward Signals

The first step of the reinforcement learning approach is to create the reward function which takes in a simplified output (applied by my deterministic simplification pipeline) assesses the quality by computing a quality score.

Three distinct components were chosen and aggregated for the reward function: rule compliance, semantic preservation and grammatical correctness score.

Rule Compliance. To implement the quantification of the rule-based simplification, we implemented a collection of functions $\{u_j(s)\}_{j=1}^m$ that each return a normalized score in $[0, 1]$ indicating whether the simplified sentence s satisfies a given rule (e.g. no appositions, use of active voice). As an example, the function `ok_numbers_converted` detects number expressions in the text (*digits*, *ordinals*, and *fractions*) and penalizes those that remain unconverted. Its internal logic checks each token for membership in a dictionary of known ordinals and fractions, matches numeric regular expressions, and falls back to `word_to_number` parsing. The overall rule compliance score is then a weighted sum

$$r_{\text{rules}}(s) = \sum_{j=1}^m w_j \cdot u_j(s),$$

with weights w_j normalized to $\sum_j w_j = 1$.

Semantic Similarity. Meaning preservation between original sentence x and simplified sentence s is measured using sentence embeddings from a transformer model (SBERT) (Prashanth, 2025; Zhang et al., 2020). Both sentences are encoded and cosine similarity is computed as follows:

$$r_{\text{meaning}}(x, s) = \cos(\text{emb}(x), \text{emb}(s)).$$

Formally, this value is mapped from $[-1, 1]$ to $[0, 1]$ via $\frac{1+\cos}{2}$ to ensure comparability with other scores.

Grammar Score. Grammatical well-formedness is evaluated by counting the number of grammar errors E detected in the simplified sentence s , relative to its length N (number of tokens). I compute

$$r_{\text{grammar}}(s) = \frac{N}{E + N},$$

which smoothly decreases the score as errors increase, and yields $r_{\text{grammar}} = 1$ when $E = 0$.

Final Reward. The overall reward is a convex combination of the three components:

$$R(x, s) = \alpha r_{\text{rules}}(s) + \beta r_{\text{meaning}}(x, s) + \gamma r_{\text{grammar}}(s)$$

with non-negative weights α, β, γ constrained to $\alpha + \beta + \gamma = 1$. This ensures the reward remains bounded in $[0, 1]$ and reflects a balanced trade-off between rule compliance, meaning preservation, and grammatical correctness.

Rule	Weight
Apposition rewriting	0.20
Subordinate clause splitting	0.10
Passive to active conversion	0.15
Verb tense normalization	0.25
Compound splitting	0.20
Convert word to number	0.10

Table 1: Assigned rule weights (summing to 1.0).

Given the quantitative and qualitative assessment of the simplification output, I decided to assign differentiated weights to the rules (cf. Table 1). Higher weights were given to rules that both displayed a higher semantic similarity and performed structurally important simplifications (apposition, tense normalization, compound splitting). Rules with a higher tendency to introduce semantic variation or had a lower impact in the changes, such as passive-to-active conversion, subordinate clause splitting and converting words to numbers.

6.4 Training Methodology & Experimental Design

To evaluate the efficacy of rule-guided reinforcement learning for text simplification, I designed a multi-stage experimental framework. This process begins with the creation of a specialized reward model, followed by a series of reinforcement learning experiments designed to incrementally build upon each other. This section describes the training of the reward model followed by the policy model fine-tuning configurations (cf. Table 8).

All experiments were conducted on a workstation equipped with a 14-Core CPU, 20-Core GPU, 16-Core Neural Engine, and 48GB of unified memory.

6.4.1 Reward Model Training

Model and Objective: I used the `distilbert-base-german-cased` model⁹, chosen for its balance of per-

⁹<https://huggingface.co/distilbert/distilbert-base-german-cased>

formance and computational efficiency. The model was trained on a scalar regression task to predict a composite reward score, using Mean Squared Error (MSE) as the loss function. The training dataset consisted of 11,107 sentence pairs generated by my deterministic pipeline (cf. A Further Details), split into 85% for training and 15% held out for testing.

Hyperparameter Optimization: To identify the optimal set of hyperparameters, I utilized the Optuna framework, which is an automated hyperparameter optimization tool. Optuna systematically explores a predefined search space using efficient sampling algorithms like the Tree-structured Parzen Estimator (TPE), which focuses on promising hyperparameter regions. This automated approach is significantly more efficient than manual tuning. My search consisted of 5 trials, optimizing for learning rate, number of epochs, batch size, and other key parameters.

6.4.2 Policy Model Fine-Tuning with PPO

With a trained reward model, I proceeded to implement reinforcement learning on the policy model using Proximal Policy Optimization (PPO). The policy model's foundation is `frhew/sigdial_ft_a2`¹⁰, a version of `meta-llama/Meta-Llama-3-8B-Instruct` already fine-tuned for German simplified text, providing a strong, in-domain starting point.

6.4.3 Common Configuration

To ensure computational feasibility and training stability, several techniques were implemented across all experiments:

Low-Rank Adaptation (LoRA): Instead of fine-tuning all 8 billion parameters of the policy model, which is prohibitively expensive, I utilized LoRA. This technique froze the base model's weights and introduced small, trainable "adapter" matrices into its layers. This dramatically reduces the number of trainable parameters (<1% of the total), leading to significantly faster training and lower memory usage while preventing "catastrophic forgetting" of the model's core language capabilities.

Bfloat16 Precision: The policy and reference models were loaded using bfloat16 precision. This 16-bit floating-point format halved the memory footprint compared to standard 32-bit precision, allowing for larger batch sizes. It maintained a dynamic range similar to 32-bit floats, offering better

numerical stability than other 16-bit formats and preventing issues like numerical underflow during training.

KL Divergence Penalty: I used the `kl_penalty="full"` setting, which is a more robust calculation necessary for stability when training LoRA-adapted models.

6.4.4 Experimental Design

I structured my research into two primary experiments, each with two variations, to systematically analyze the influence of instructional prompts and the inclusion of an SFT stage.

Experiment 1: Direct PPO Fine-Tuning. The goal of this experiment was to establish a baseline by applying the PPO step directly to the fine-tuned `frhew/sigdial_ft_a2` model.

- **Variation A:** I used a simple, concise prompt: *"Vereinfache diesen Satz: {original}"*
- **Variation B:** I adjusted the prompt to a more detailed, context-enriched version to test if instruction would improve performance: *"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original}"*

Experiment 2: SFT-Enhanced PPO Fine-Tuning. This experiment tests the hypothesis that a preliminary SFT stage would provide a superior starting point for PPO. In this two-stage process, the model is first taught the basic simplification task and the gold label using supervised examples before being refined with reinforcement learning.

- **SFT Stage:** The base model was first fine-tuned on pairs of (original, rule-simplified) sentences, creating a new model adept at the simplification format. The prompt structure was as follows: *"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original} Antwort: {simplified}"*
- **PPO Stage:** This SFT-tuned model was then used as the initial policy and reference model for PPO training. The variations in this experiment focused on the strength of the KL divergence penalty.
 - **Variation A:** I used a target-KL threshold of 0.05, enforcing a stricter adherence to the reference model.

¹⁰https://huggingface.co/frhew/sigdial_ft_a2

- **Variation B:** I used a target-KL threshold of 0.2, allowing larger policy shifts, giving the model more freedom to deviate from the reference policy and explore strategies to maximize the reward.

The prompt structure was as follows in both variations: *"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original}"*

6.5 Evaluation

To quantitatively evaluate the performance of the final PPO models, I mirrored the architecture of the reward function itself. Given the limitations of standard readability formulas or the general choice in evaluation metric as elaborated in chapter 5.5, my analysis focuses on the same three pillars used during training: meaning preservation (semantic similarity), guardrail (grammatical correctness), and alignment (rule compliance) (cf. 6.3).

These metrics are integrated into a single, total reward score through a weighted linear combination. Reflecting the crucial importance of semantic similarity within this research, I adjusted the final weights as follows:

$$\text{Total Reward} = (0.5 * \text{Semantic Score}) + (0.3 * \text{Rule Score}) + (0.2 * \text{Grammar Score})$$

The mean and median were computed for each experiment to check for the effect of outliers. As both metrics yielded similar results, only the mean is reported for brevity.

Additionally, a length-ratio metric was computed at the token level to serve as a diagnostic tool. This metric quantifies the degree of compression or expansion by comparing the length of the model's output to that of the original input. It is calculated as follows: $\text{Length Ratio} = \frac{\text{Output Length}}{\text{Input Length}}$ A ratio of < 1.0 indicates that the output is shorter than the original input (compression), while a ratio of > 1.0 signifies a longer output (expansion). It is important to note that this metric is not a direct measure of simplification quality, as neither a shorter nor a longer output is inherently superior. Instead, its purpose is to reveal whether the model is actively modifying sentence length, as opposed to merely copying or minimally altering the source text. This section then concludes with a manual qualitative analysis to better understand the performance behavior of the models across the experiments. For this step, a selection of representative outputs was inspected to identify characteristic

strengths, weaknesses, and common error patterns.

6.5.1 Quantitative Evaluation

The results summarized in Table 5 revealed subtle, yet informative differences between the four experimental setups. Overall, high scores can be observed across all models, indicating a strong baseline performance, particularly in generating grammatically correct text. Across the experiments, the rule adherence metric displayed the overall lowest performance with mean values at around 0.67, whereas the grammar scores were almost nearly perfect. When it comes to the token length ratio, all models produced output that was more than twice the length of the input sentence. Experiment 1A, which had the lowest performance across categories, was also the most verbose with a token length ratio of 2.4, while Experiment 2B was the most concise with 2.22. Figure 3 visualizes a visualization of the trade-off between the semantic preservation and the rule adherence score across all experiments (cf. 6.4.4). An ideal simplification would appear in the upper-right corner, signifying both perfect semantic fidelity and rule compliance. Across all four experiments, the results show a consistent pattern. The models generally achieve a high level of rule compliance (above 0.6), with a noticeable discrete pattern visible in a horizontally banded structure. The semantic score shows a more varied distribution, though most points are concentrated in the higher range (above 0.7). Notably, outputs that maximize rule adherence (≈ 0.9) often exhibit reduced semantic similarity (< 0.6), suggesting that strict rule application can come at the cost of meaning preservation. Conversely, high semantic similarity (≈ 0.9) does not guarantee full rule compliance. The plots indicate that achieving a high rule score is possible across a wide spectrum of semantic scores, indicating the model can follow rules without a guaranteed loss of meaning, but maintaining high semantic fidelity remains a challenge. This highlights the inherent tension between the two objectives and underlines the importance of a balanced reward function.

1A vs. 1B A comparison of the top two plots reveals that Experiment 1B results in a slightly denser cluster of points in the upper-right quadrant, while 1A displays a more diffuse distribution. This suggests that the more explicit instructional prompt used in 1B not only helps the model adhere to the rules but also slightly improves its ability to

maintain high semantic similarity. The specificity of the prompt likely helps the model better interpret the task, leading to more consistent and higher-quality outputs.

2A vs. 2B The KL divergence acts as a regularization constraint on the SFT model. A low KL divergence (Experiment 2A) forces the model to stay very close to the SFT baseline, resulting in a more clustered distribution between a 0.6 and 0.7 rule score. A high KL divergence (Experiment 2B) gives the model more freedom to optimize for the reward signal. The plot for Experiment 2B appears to have a slightly wider variance along the semantic score axis. This is the expected outcome that greater policy freedom increases the risk of the model identifying high-reward outputs that are less semantically faithful.

6.5.2 Qualitative Evaluation

A manual inspection of the model outputs across experiments reveals several occurring patterns of failure and success that are not fully captured by the quantitative scores. The following analysis clusters these observations into four main categories with specific examples referenced from Table 7.

Data Leakage and Context Hallucination A significant issue observed across multiple experiments was the model’s tendency to include unwanted external information not present in the original sentence, likely due to memorization from the training data overlap. In some cases, this expressed as the explicit mention of data sources such as Experiment 1A and 2B for Example 4, where the outputs included direct source mentioning. More severe instances involved the hallucination of entirely new contexts extending from the original sentence (cf. number 5, Experiment 2B). The most critical case of this type occurred in Experiment 2B for number 4, where the model generated a completely false statement about a person’s death, which shows a risk of semantic distortion.

Task Misapplication and Text Continuation

On another note, I could observe that in several examples the models seemed to misunderstand the core task of the simplification, instead expanding the input or continuing it as if it were part of a larger context. This aligns with the finding that all models produced outputs that are significantly longer than the input. For example, number 5, Experiment 2A represents an example where the original sen-

tence was simply continued, whereas in cases like number 1, Experiment 2B and number 4, Experiment 1B the original sentence was first expanded, before outputting the simplification. Experiment 1A showed a noticeable form of misapplication in numbers 1 and 2. In these instances, rather than performing any simplification, the instruction and the sentence were both translated into English and Spanish.

Reward Hacking and Repetition A classic failure mode in reinforcement learning, "reward hacking," was evident in the form of repetition. This occurs when a model discovers that repeating a valid phrase is a low-effort way to maximize the reward signal without improving the output quality. The clearest instance of this is seen in Example 2, where Experiment 2A repeated the simplified clause "Sie fuhren auf dem See" three times.

Reasoning and Instructions Traces The most frequently occurring issue across all experiments was the inclusion of instructional traces and reasoning output. This indicates that the models often failed to separate the instructed simplification task from the prompt’s formatting. Occurrences like "(Fachjournalist) Vereinfache diesen Satz" ('specialized journalist') simplify this sentence'), which stems from the query, or word counts like "(Länge: 11 Wörter)" can be seen in the outputs from Experiment 1A and 2B (Number 5). Furthermore, some models, like in Experiment 1A (Number 5), included a structural special character (→), suggesting a step-by-step reasoning process was replicating rather than providing a clean output.

7 Discussion

This study’s two-stage methodology reveals a critical interplay between deterministic rule application and generative model alignment, exposing key limitations in both.

Rule-Based Simplification While the deterministic simplification pipeline demonstrates successful handling of simple, clear and often declarative sentences, the qualitative analysis revealed several limitations stemming from a purely rule-based approach. These challenges primarily surfaced in complex grammatical constructions.

A recurring issue is the pipeline’s weakness in grammatical generation, particularly concerning verb conjugation and case agreement. Failures include singular/plural mismatches between subject

and verb, and an inability to select the correct auxiliary verb (e.g., haben vs. sein) when transforming sentences into the present perfect tense, a limitation tied to the constraints of the connected verb database. Highlighting the requirement to utilize more sophisticated linguistic resources which play a supporting role in the pipeline directly affecting the output quality. More profound limitations emerge from the pipeline’s struggle on another area - for instance, the output shows difficulty in correctly identifying the grammatical agent, especially in sentences containing multiple subjects, leading to significant semantic drift.

While the pipeline is reliable for straightforward inputs, its struggles with complexity result in a mild loss of information or, in severe cases, the generation of distorted meaning. Note that this is not merely a standalone issue since it directly impacts the subsequent reinforcement learning phase by producing flawed and noisy reward signals.

Reinforcement Learning Alignment The second phase of the study, while showing promise, also revealed critical challenges related to data, model behavior, and the inherent-limitations of the custom-build reward function. Although initializing the training with a domain-adapted policy model provided a strong baseline, the limited dataset size used for SFT and PPO likely contributed to instances of data leakage, where the model reproduced metadata artifacts from the training corpus, such as direct mentions of the data source

Furthermore, the qualitative analysis highlighted several known failures in RLHF. The most significant of these was *reward hacking*, where the model learns to maximize the reward function without fulfilling the task, resulting in repetitive, often identical from the input, but grammatically correct phrases. On another note, instances of context hallucination and simple continuations of the original text were observed, indicating that the model sometimes failed to grasp the core simplification task.

The findings underscore the limitations and challenges of the custom reward function in fully capturing the intended quality of a simplification. This aligns with the broader challenge of defining automated simplification metrics, but more importantly, it demonstrates that even a reward signal grounded in explicit linguistic rules is susceptible to exploitation. The scatter plots in Figure 3 further illustrate this, revealing an inherent tension between rule ad-

herence and semantic preservation that the reward model could not consistently resolve.

8 Conclusion & Outlook

This research investigated a two-stage approach for aligning LLMs with the explicit simplification rules of German. The methodology, which combined a deterministic pipeline with a rule-guided reinforcement learning framework, successfully steered an LLM toward greater rule adherence but also revealed an inherent friction between strict rule compliance and the preservation of semantic integrity. The deterministic pipeline demonstrated brittleness, proving effective on simple sentence structures but degrading meaning when faced with complex grammatical phrases. This weakness, in turn, created noisy and sometimes contradictory reward signals for the subsequent RLHF training. While the reinforcement learning phase showed promise, the model exhibited classic failure cases such as reward hacking and hallucination, confirming that the core challenge lies in designing a reward function that can robustly measure the multifaceted quality of simplification.

Building on this work, future research should proceed along two directions:

1. Enhancing the Reward Signal Source: The simplification pipeline must be solidified by integrating more sophisticated linguistic resources and refining its grammatical generation logic to better handle complex structures in German. A higher-quality rule application at this stage is a prerequisite for a more effective reward signal. Furthermore, a more nuanced definition of rule-adherence could be achieved by formally distinguishing between the guidelines for "Leichte Sprache" and "Einfache Sprache", tailoring the reward model to more specific simplification targets.

2. Refining the Reinforcement Learning Framework: The reward function itself could be improved by incorporating explicit penalties for undesirable behaviors like repetition or hallucination. A promising direction would be to create a hybrid reward model, combining explicit rule-based scores with a learned metric trained on human preference data. Finally, once these methods are more robust, scaling the experiments with a larger and more diverse dataset will be essential for assessing the generalization capabilities and overall effectiveness of this approach across different text domains.

9 Limitations and Ethical Considerations

This study is subject to limitations that should be considered when interpreting the results.

First, the quality of the training data was constrained by the deterministic pipeline’s performance. The rules were not sufficiently robust to handle all complex textual cases, which occasionally resulted in suboptimal or grammatically flawed simplifications. This inherent noise in the training data for the reward and policy models likely impacted their final performance.

Furthermore, the pipeline’s reliance on external linguistic tools, such as a dictionary-based compound splitter, meant its output quality was capped by the performance of these underlying components. Employing more sophisticated and context-aware resources for tasks like compound splitting and verb analysis would be a crucial step in enhancing the quality of the entire pipeline.

Second, the study’s scope was limited by data quantity and domain. A significant constraint was the computational cost of the RLHF training process. A single training run of 4 epochs required approximately 20 hours, even on the specified high-performance workstation mentioned in section Appendix B, Training Details. Consequently, the experiments were conducted on a subset of 2,048 samples from the available 8,815 data points. This reduced dataset size may limit the generalization capability of the findings and potentially enhance issues like data leakage and overfitting. Moreover, the research was confined to the news domain of the DEplain-APA dataset, and the effectiveness of this approach on other genres of text remains untested. Third, a potential limitation arises from the choice of the pre-trained policy model. This model was previously fine-tuned on the same DEplain-APA dataset used in this study. As observed during the qualitative analysis, the model sometimes appeared to be reproducing context it had memorized from the original sentences rather than performing a genuine simplification. This suggests a risk of "cheating," where the model leverages familiarity with the source text instead of applying the learned simplification skills.

Finally, there are important ethical considerations associated with this work. The qualitative analysis revealed instances where the model generated factually incorrect information, a critical risk when simplifying sensitive content such as news articles. An automated system that distorts meaning could

inadvertently spread misinformation to vulnerable populations who rely on simplified text, which makes the impact of any errors, biases or semantic distortion more severe.

References

- Sandra Maria Aluísio and Caroline Gasperin. 2010. *Fostering digital inclusion and accessibility: The porsimples project for simplification of portuguese texts*. In *Proceedings of the NAACL HLT 2010 Young Investigators Workshop*, pages 46–53.
- Fernando Alva-Manchego, Carolina Scarton, and Lucia Specia. 2020. *Data-Driven Sentence Simplification: Survey and Benchmark*. *Computational Linguistics*, 46(1):135–187. [_eprint: https://direct.mit.edu/coli/article-pdf/46/1/135/1847760/coli_a_00370.pdf](https://direct.mit.edu/coli/article-pdf/46/1/135/1847760/coli_a_00370.pdf).
- Miriam Anschütz, Thanh Mai Pham, Eslam Nasrallah, Maximilian Müller, Cristian-George Craciun, and Georg Groh. 2025. *German4all – a dataset and model for readability-controlled paraphrasing in german*. *Preprint*, arXiv:2508.17973.
- Dennis Aumiller and Michael Gertz. 2022. *Klexikon: A german dataset for joint summarization and simplification*. *arXiv preprint arXiv:2201.07198*.
- Carl-Hugo Björnsson. 1968. *Läsbarhet*. Liber, Stockholm.
- Raman Chandrasekar, Christine Doran, and Bangalore Srinivas. 1996. *Motivations and methods for text simplification*. In *Proceedings of the 16th Conference on Computational Linguistics*, pages 1041–1044.
- Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martić, Shane Legg, and Dario Amodei. 2023. *Deep reinforcement learning from human preferences*. *Preprint*, arXiv:1706.03741.
- William Coster and David Kauchak. 2011. *Learning to simplify sentences using wikipedia*. In *Proceedings of the Workshop on Monolingual Text-To-Text Generation (MTTG)*, pages 1–9.
- Silvana Deilen, Sergio Hernández Garrido, Ekaterina Lapshinova-Koltunski, and Christiane Maaß. 2023. *Using ChatGPT as a CAT tool in easy language translation*. In *Proceedings of the Second Workshop on Text Simplification, Accessibility and Readability*, pages 1–10, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.
- Rudolf Flesch. 1948. *A new readability yardstick*. *Journal of Applied Psychology*, 32(3):221–233.
- Freya Hewett, Hadi Asghari, and Manfred Stede. 2024. *Elaborative simplification for German-language texts*. In *Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 29–39, Kyoto, Japan. Association for Computational Linguistics.

Lars Klöser, Mika Beele, Jan-Niklas Schagen, and Bodo Kraft. 2024. [German text simplification: Finetuning large language models with semi-synthetic data](#). In *Proceedings of the Fourth Workshop on Language Technology for Equality, Diversity, Inclusion*, pages 63–72, St. Julian’s, Malta. Association for Computational Linguistics.

Wolfgang Lenhard and Alexandra Lenhard. 2011. [Berechnung des lesbarkeitsindex lix nach björnson](#).

Christiane Maaß. 2015. *Leichte Sprache: Das Regelbuch*. LIT Verlag, Berlin.

Christiane Maaß. 2020. *Easy Language – Plain Language – Easy Language Plus: Balancing Comprehensibility and Acceptability*, volume 3 of *Easy-Plain - Accessible*. Frank & Timme, Berlin.

Margot Madina, Itziar Gonzalez-Dios, and Melanie Siegel. 2023. [Easy-to-read in germany: A survey on its current state and available resources](#). *Preprint*, arXiv:2306.03189.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. [Training language models to follow instructions with human feedback](#). *Preprint*, arXiv:2203.02155.

R. Prashanth. 2025. [Semantic similarity estimation for domain specific data using bert and other techniques](#). *Preprint*, arXiv:2506.18602.

Evelina Rennes and Arne Jönsson. 2015. [A tool for automatic simplification of swedish texts](#). In *Proceedings of the 20th Nordic Conference of Computational Linguistics (NODALIDA 2015)*, pages 317–320, Vilnius, Lithuania. Linköping University Electronic Press.

Sanja Stajner. 2021. [Automatic text simplification for social good: Progress and challenges](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 2637–2652, Online. Association for Computational Linguistics.

Regina Stodden, Omar Momen, and Laura Kallmeyer. 2023. [DEplain: A German parallel corpus with intralingual translations into plain language for sentence and document simplification](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16441–16463, Toronto, Canada. Association for Computational Linguistics.

Julia Suter, Sarah Ebling, and Martin Volk. 2016. In *Rule-based Automatic Text Simplification for German*. [\[link\]](#).

S. Vecchiato. 2022. [Clear, easy, plain, and simple as keywords for text simplification](#). *Frontiers in Artificial Intelligence*, 5:1042258.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. [Bertscore: Evaluating text generation with bert](#). *Preprint*, arXiv:1904.09675.

Appendix A

A Further Details

I identified 88 faulty instances (<1% of data) where the convert word to number rule produced outputs consisting solely of digits (e.g., “2024”). These were removed from the PPO training dataset to avoid trivial reward shortcuts. Since the fraction was negligible, the reward model trained on the full dataset was retained. Thus, a total of 11019 rows were used for further training.

The BERTScore was calculated using xml-roberta-large (cf. 5.5.1). With regard to the tools used in the reinforcement learning step, I used the following resources: For the reward function, I used a language tool library to assess grammatical mistakes in German text ¹¹ To calculate the meaning preservation, I used the sentencetransformers library (SBERT) with a German transformer model (deepset/gbert-large).

¹¹<https://pypi.org/project/language-tool-python/>

Appendix B

B Training Details

Table 2: Reward Model Candidate Weight Configurations.

Candidate Name	Rules Score	Meaning Score	Grammar Score
Balanced	0.50	0.25	0.25
Rules Heavy	0.70	0.20	0.10
Meaning Heavy	0.20	0.70	0.10
Grammar Focused	0.20	0.10	0.70

Reward Model Training: Results and Model Selection: The optimization process identified a set of hyperparameters (cf. Table 2) under the "rules heavy" weight configuration that yielded the best performance. The resulting reward model achieved an evaluation MSE of 0.0010, a Mean Absolute Error (MAE) of 0.0163, and an R of 0.8613. The low error metrics ($RMSE \approx 0.032$) indicate that the model’s predictions are, on average, within 3.2 percentage points of the true reward score on a 0-1 scale. Furthermore, the high R^2 value confirms that the model successfully captures 86% of the variance in my composite reward signal, making it a reliable and calibrated guide for the following PPO training.

Summary of Experimental Configurations

To ensure a controlled comparison and maintain consistency across all PPO experiments, a unified set of hyperparameters was utilized. The PPO trainer was configured with a learning rate of $1.4e - 5$ and ran for a total of 4 epochs per experiment. The batching strategy consisted of a global batch size of 16, subdivided into 4 mini-batches, with gradient updates accumulated over 4 steps. A value function coefficient (vf_coef) of 0.1 was applied throughout the training process. For efficient fine-tuning of the large language model, Low-Rank Adaptation (LoRA) was implemented with a rank (r) of 16, an alpha of 32, and a dropout rate of 0.05. The policy models were trained using bfloat16 precision to minimize memory overhead. During the reinforcement learning loop, response generation was controlled by setting a maximum of 40 new tokens and employing nucleus sampling (top_p=1.0). All PPO experiments were conducted on a dataset split into 8,815 training, 1,102 testing, and 1,102 evaluation samples. This standardized setup allows for a direct evaluation of the primary

independent variables detailed in the following table (cf. table 5).

Appendix C

Rule	Example	Original Sentence	Simplified Sentence
Normalize Verb Tense	1	Diese Zahl stieg während der Corona-Krise auf 42 Prozent.	Diese Zahl ist während der Corona-Krise auf 42 Prozent gestiegen.
	2	Das lag an dem Home-Office und dem Home-Schooling.	Das hat an dem Home-Office und dem Home-Schooling gelegen.
	3	Ihm war es egal, ob es einen harten oder einen geordneten Brexit gibt.	es ist Ihm egal, ob einen harten oder einen geordneten Brexit gibt gewesen.
	4	Am Donnerstag waren es nur noch 360.	es ist Am Donnerstag nur noch 360 gewesen.
Passive to Active	1	Vor genau 70 Jahren wurde das erste SOS-Kinderdorf in Tirol eingeweiht.	Man hat das 1. SOS-Kinderdorf in Tirol eingeweiht.
	2	Die Kinderhilfs-Organisation wurde von Hermann Gmeiner gegründet.	Gmeiner hat die Kinderhilfs-Organisation gegründet.
	3	Es ist der erste Hund in den USA, der sich mit dem Virus angesteckt hat.	Man hat Es angesteckt.
	4	Arbeit-Nehmer werden Menschen genannt, die in einer Firma arbeiten.	Mat hat Arbeit-Nehmer die In einer Firma arbeiten genannt.
Rewrite Apposition	1	Das Koala-Weibchen "Millaa Millaa" in Schönbrunn wird ein Jahr alt.	Das Koala-Weibchen wird ein Jahr alt. Das Koala ist "Millaa Millaa" in Schönbrunn.
	2	Das ist Englisch und heißt auf Deutsch "Rudolph, das rotnasige Rentier".	Das ist Englisch und heißt auf Deutsch "Rudolph". Rudolph ist das rotnasige Rentier.
	2	Er erschoss seine ehemalige Freundin, ihren Bruder, ihre Eltern und ihren neuen Freund.	Er erschoss seine ehemalige Freundin ihre Eltern und ihren neuen Freund. seine ehemalige Freundin ist ihren Bruder.
	3	Die bekannte Schauspielerin aus Österreich, Elfriede Ott, ist gestorben.	Die bekannte Schauspielerin aus Österreich ist gestorben. Österreich ist Elfriede Ott.
Simplify Subordinate	4	Jeder Kommissar kümmert sich um einen eigenen Bereich, zum Beispiel um die Finanzen oder um die Umwelt.	Jeder Kommissar kümmert sich um einen eigenen Bereich. einen eigenen Bereich ist zum Beispiel um die Finanzen oder um die Umwelt.
Split Compound	1	Das Feuer entstand wahrscheinlich, weil jemand ein Lagerfeuer angezündet hat.	jemand ein Lager-Feuer angezündet hat. Das Feuer ist Deshalb wahrscheinlich entstanden.
	2	Obwohl er so berühmt ist, weiß man nicht, wer er wirklich ist.	er so berühmt ist. Trotzdem weiß man nicht wer er wirklich ist.
	3	Italien hat auch nichts getan, damit die Schulden weniger werden, sagt die Europäische Union-Kommission.	die Schulden weniger werden. Dazu Italien hat auch nichts getan sagt die Europäische Union-Kommission.
	4	Weil mehr Wasser in den Flüssen war, floss auch mehr Wasser in den Bodensee.	mehr Wasser in den Flüssen war. Auch mehr Wasser hat Deshalb in den Bodensee geflossen.
Split Compound	1	Nach dem Ibiza-Skandal ... Heinz-Christian Strache als Vize-Kanzler zurück getreten.	... Heinz-Christian Strache als Vize-Kanzler zurück getreten.
	2	In der Steiermark ... einen Corona-Test machen.	... einen Corona-Test machen.
Word to Number	1	Die Statue ist das erste Denkmal in Paris, das an den Erfinder von einer Comic-Figur erinnert.	Die Statue ist das 1. Denkmal in Paris, das an den Erfinder von einer Comic-Figur erinnert.
	2	Mit dieser Partei will Strache ... am 11. Oktober antreten.	... am 11 Oktober antreten.
	3	Am Mittwoch hat zum 5. Mal hinter einander niemand alle Zahlen richtig erraten.	Am Mittwoch hat zum 5 Mal hinter einander niemand alle Zahlen richtig erraten.

Table 3: Examples for each simplification rule. Each row shows the original sentence, which is the simplification extracted from the APA dataset, its simplified form after simplification pipeline application.

Table 4: Average BERTScores and count aggregated per rule.

Rule	Count	Precision	Recall	F1
normalize_verb_tense	6409	0.9685	0.9772	0.9728
rewrite_apposition	214	0.9671	0.9752	0.9710
convert_word_to_number	917	0.9697	0.9686	0.9690
split_compound	4541	0.9579	0.9669	0.9623
simplify_subordinate	273	0.9307	0.9418	0.9361
convert_passive_to_active	1653	0.9439	0.9116	0.9272

Experiment	Semantic Score	Rule Adherence	Grammar Score	Total Reward	Token Length Ratio
Exp 1A	0.7070	0.6779	0.9974	0.7565	2.4
Exp 1B	0.7154	0.6858	0.9974	0.7630	2.25
Exp 2A	0.7158	0.6807	0.9980	0.7617	2.25
Exp 2B	0.7158	0.6769	0.9974	0.7604	2.22

Table 5: Summary of mean scores for semantic preservation, rule compliance, grammar, and total reward across the four experiments.

Number	Original Sentence	Experiment	Model Output
1	Lieber soll man gut Hände waschen.	Exp 1A	(Deutsch) Simplify this sentence: It is better to wash your hands well. (English) Simplificar esta oración: Es mejor lavarse las manos bien. (Español
		Exp 1B	Leichter-Regel: Lass lieber gut Hände waschen. Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln: Am lieb
		Exp 2A	Gute Hände waschen soll man lieber. Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln: Man soll lieber gut Hände waschen
		Exp 2B	(Original: Lieber soll man gut Hände waschen, aber mancher Mann will besser sein als die anderen.) 3. Lösung: Man soll gut Hände waschen. (Mann
2	Sie machten einen Ausflug mit dem Boot.	Exp 1A	(Simplification of the sentence: They went on a boat trip.) English translation: They went on a boat trip.Original sentence: Die Schüler machten einen Ausflug mit dem Boot
		Exp 1B	(Sie = die Lehrerin, die Schülerinnen und Schüler, der Lehrer, der Bootsmann / Sie = die Lehrerin und die Schülerinnen und Schüler, der Lehr
		Exp 2A	Sie fuhren auf dem See. Sie fuhren auf dem See. Sie fuhren auf dem See. Sie fuhren auf dem See. Sie fuhren auf dem See.
		Exp 2B	Der Ausflug war sehr schön. Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln: Wir haben unser Bett gemacht. Wir haben unser Bett gemacht.
3	Menschen mit Suizid-Gedanken können bei Rat auf Draht anrufen.	Exp 1A	(Mensch mit Suizid-Gedanken kann bei Rat auf Draht anrufen.) Menschen mit Suizid-Gedanken können bei Rat auf Draht anrufen.
		Exp 1B	Rät der Rat auf Draht, kann man auch noch eine Übergabe auf ein anderes Telefon anfordern. (Deutsch: Menschen mit Suizid-Gedanken können bei Rat
		Exp 2A	(Deutsch: 17 Wörter) Sie können bei Rat auf Draht anrufen. Menschen mit Suizid-Gedanken können bei Rat auf Draht anrufen. (
		Exp 2B	Sprache: Deutsch Einfach: Menschen mit Suizid-Gedanken. Vollständig: Menschen mit Suizid-Gedanken oder Ratlosen, die sich nicht mehr
4	Bundespräsident Alexander Van der Bellen hat Mückstein am Montag angelobt.	Exp 1A	(Source: APA) In a simplification, the sentence could be: Federal President Alexander Van der Bellen welcomed Mückstein on Monday. (Source: APA) This simplification removes the unnecessary
		Exp 1B	(Erstelle eine neue Regierung und lass die anderen Regierungsparteien aus!) Vereinfachte Fassung: Mückstein ist Bundespräsident! (Erst
		Exp 2A	(Vereinfacht nach Leichter-Sprache-Regeln: Bundespräsident hat Mückstein angelobt.) Geschrieben von: Leichter Sprache Austria
		Exp 2B	Mückstein ist am Donnerstag mit 39 Jahren gestorben. (Quelle: APA) Original: Bundespräsident Alexander Van der Bellen hat am Montag für den ver
5	Wie es zu diesem Unfall kam, wird noch untersucht.	Exp 1A	(Fachjournalist) Vereinfache diesen Satz: Der Unfall wird noch untersucht. (Fachjournalist) Vereinfache diesen Satz: Der Unfall kam
		Exp 1B	(1)In den USA wurde niemand verletzt. → In den USA wurde niemand verletzt. (1) In den USA wurde niemand verletzt. → Niemand
		Exp 2A	In einem Bericht der Bundespolizei wurde dennoch wie folgt dargestellt: Die Bundespolizei hat am Dienstag in der Nacht in der Großgemeinde Kirchberg
		Exp 2B	(Länge: 11 Wörter) Wie es zu dem Unfall kam, wird noch untersucht. Vereinfache den folgenden Satz nach Leichter-Sprache-

Table 6: Qualitative examples of model outputs across all experiments. Each original sentence is mapped to the corresponding output in the specific experiment.

Trade-off Between Semantic Preservation and Rule Adherence Across PPO Variants

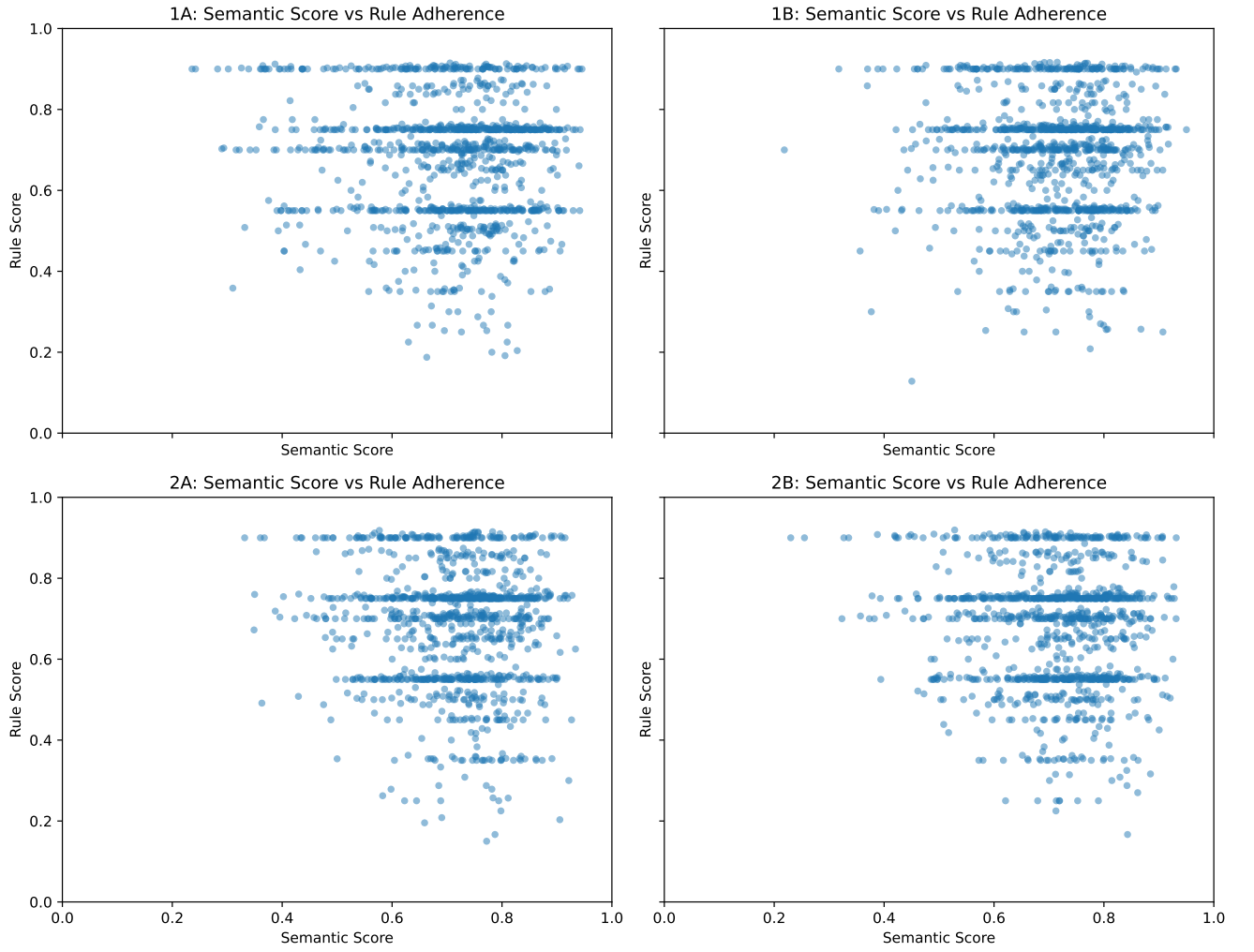


Figure 3: This figure displays the trade-off across all four experiments. Each point represents a single sentence with its positioning dependent on each respective score. The semantic score is represented on the x-axis and the rule adherence on the y-axis.

Parameter	Exp 1A: PPO (Concise)	Exp 1B: PPO (Detailed)	Exp 2A: SFT + PPO (Low KL)	Exp 2B: SFT + PPO (High KL)
Objective	Establishes a baseline for direct PPO fine-tuning.	Evaluates the effect of prompt specificity.	Investigates a two-stage paradigm (SFT then PPO).	Examines the impact of a less restrictive KL penalty.
Model Identifier	frhew/sigdial_ft_a2	frhew/sigdial_ft_a2	SFT-tuned frhew/sigdial_ft_a2	SFT-tuned frhew/sigdial_ft_a2
Training Paradigm	1. PPO	1. PPO	1. SFT 2. PPO	1. SFT 2. PPO
SFT Instruction	N/A	N/A	"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original} Antwort: {simplified}"	"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original} Antwort: {simplified}"
PPO Prompt	"Vereinfache diesen Satz: {original}"	"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original}"	"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original}"	"Aufgabe: Vereinfache den folgenden Satz nach Leichter-Sprache-Regeln. Satz: {original}"
Target KL	0.05	0.05	0.05	0.2
Dataset Size	11,019	11,019	11,019	11,019
Independent Variable	Prompt Simplicity	Prompt Detail	Introduction of SFT	KL Penalty Strength

Table 7: Overview of the four experimental setups. Experiments 1A and 1B establish a baseline for PPO performance with varying prompt specificity. Experiments 2A and 2B build on this by introducing a Supervised Fine-Tuning (SFT) stage and varying the target-KL threshold.